

Mix It Up. On Purpose.

Interleaving · Kornell & Bjork · Rohrer & Taylor

"I studied one topic for an hour and aced the practice problems.
Then the exam asked me to tell two similar concepts apart."

— Blocked practice builds fluency within a category. Interleaving builds discrimination between them.

The most natural way to study is to finish one topic completely before moving to the next. This is **blocked practice**. It feels efficient, produces fast short-term gains, and in the research it consistently underperforms **interleaved practice** — where you mix topics deliberately — on every test that matters: delayed recall, transfer, and discrimination between similar concepts.

The Research

Kornell and Bjork (2008) had participants learn to identify paintings by artist. Blocked learners studied all paintings by each artist in sequence. Interleaved learners saw paintings mixed randomly. Blocked learners performed better during practice. Interleaved learners performed dramatically better on the final test — and most participants mistakenly believed blocked practice had worked better for them.

Rohrer and Taylor (2007) replicated this in mathematics: students who practiced problems in mixed order performed **43% better** on a delayed test than students who practiced in blocked order, with equivalent study time.

+43%

better delayed test performance with interleaving vs. blocked (Rohrer & Taylor, 2007)

feels ↓

Interleaving feels harder during study — which is precisely why most students avoid it

which?

Interleaving teaches your brain to identify WHICH strategy applies — the real-world skill

Why It Works: The Discrimination Problem

When you study blocked, your brain doesn't need to identify **which category** the current item belongs to — the category is given. In an exam, the category is not given. You have to figure out which concept applies before you can use what you know. Interleaving forces your brain to solve this categorization problem on every item during practice.

The "Which one is this?" effect: Blocked practice answers "How do I handle type A?" Interleaved practice answers "Is this type A or type B — and how do I tell?" The second question is what exams ask. It's what professional practice requires.

What This Looks Like

Blocked (less effective)	Interleaved (more effective)
Study all of Chapter 3, then all of Chapter 4	Mix questions from all chapters in a single session
Practice 10 problems of type A, then 10 of type B	Mix type A, B, and C problems randomly — must identify type before solving
Review one concept group until it feels solid	Return to earlier concepts mixed with new ones — forces re-discrimination

Practice quiz design: A randomized quiz that mixes all topics without warning is not harder than blocked study — it's more accurate training for exam conditions. The discomfort of not knowing what category is coming is the mechanism, not a flaw.

Flashcard shuffle: Shuffle completely before every session. Never study in the same order twice. If you find yourself predicting what comes next, you're getting blocked-practice benefits when you need interleaving ones.

The confidence trap: Blocked practice feels better because differences within a category are obvious. Interleaved practice feels harder because differences across categories are subtle. The harder feeling produces the better exam score.

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